

785. Is Graph Bipartite?

Solved

Medium

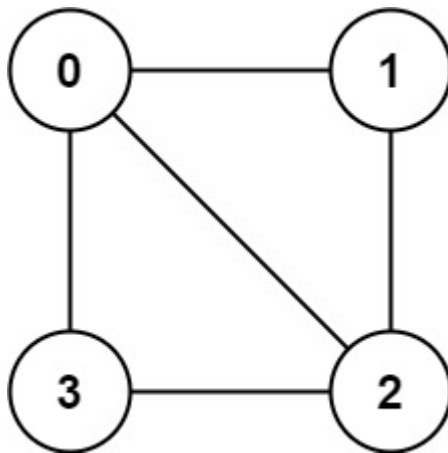
There is an **undirected** graph with n nodes, where each node is numbered between 0 and $n - 1$. You are given a 2D array `graph`, where `graph[u]` is an array of nodes that node u is adjacent to. More formally, for each v in `graph[u]`, there is an undirected edge between node u and node v . The graph has the following properties:

- There are no self-edges (`graph[u]` does not contain u).
- There are no parallel edges (`graph[u]` does not contain duplicate values).
- If v is in `graph[u]`, then u is in `graph[v]` (the graph is undirected).
- The graph may not be connected, meaning there may be two nodes u and v such that there is no path between them.

A graph is **bipartite** if the nodes can be partitioned into two independent sets A and B such that **every** edge in the graph connects a node in set A and a node in set B .

Return true *if and only if* it is **bipartite**.

Example 1:

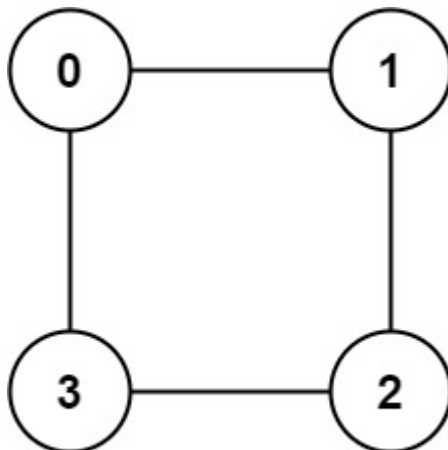


Input: `graph = [[1,2,3],[0,2],[0,1,3],[0,2]]`

Output: false

Explanation: There is no way to partition the nodes into two independent sets such that every edge connects a node in one and a node in the other.

Example 2:



Input: `graph = [[1,3],[0,2],[1,3],[0,2]]`

Output: true

Explanation: We can partition the nodes into two sets: $\{0, 2\}$ and $\{1, 3\}$.